Timo Probst

Research Assistant
Chair of Computer Graphics, Department for Computer Science
University of Freiburg, Germany

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Education

Ph.D. Student University of Freiburg, Germany
Chair of Computer Graphics, Department of Computer Science *2021 – ...*
Supervisor: Matthias Teschner

M.Sc. in Computer Science University of Freiburg, Germany
Graduated with honors. *2019 – 2021*
Thesis title: Global Contact Handling and Dry Friction for Particle-based Rigid Bodies.

B.Sc. in Computer Science University of Freiburg, Germany
Thesis title: Matrix-free PPE Solver for SPH Fluids *2015 – 2019*

Employment

Research Assistant University of Freiburg, Germany
Chair of Computer Graphics, Department of Computer Science *2021 – ...*

Student Assistant University of Freiburg, Germany
Chair of Computer Graphics, Department of Computer Science *2019 – 2021*

Tutor University of Freiburg, Germany
Python for Biologists, Faculty of Biology *2019*

Publications

[1] Timo Probst and Matthias Teschner. "Unified Pressure, Surface Tension and Friction for SPH Fluids". In: *ACM Trans. Graph.* 44.1 (Dec. 2024). ISSN: 0730-0301. DOI: 10.1145/3708034.

[2] Timo Probst and Matthias Teschner. "Monolithic Friction and Contact Handling for Rigid Bodies and Fluids Using SPH". In: *Computer Graphics Forum* 42.1 (2023), pp. 155–179. DOI: 10.1111/cgf.14727.

Invited Talks

SIGGRAPH 2025 Vancouver, Canada
Unified Pressure, Surface Tension and Friction for SPH Fluids *August 2025*

FIFTY2 Freiburg, Germany
Simulating Water Droplets with SPH *December 2024*
Friction and Contact Handling for Rigid Bodies and Fluids Using SPH *March 2023*

Eurographics 2023 Saarbrücken, Germany
Monolithic Friction and Contact Handling for Rigid Bodies and Fluids Using SPH *May 2023*

Reviews

ACM SIGGRAPH	*2025, 2026*
ACM SIGGRAPH Asia	*2026*
IEEE Transactions on Visualization and Computer Graphics	*2023, 2025, 2026*
Eurographics	*2026*
ACM SIGGRAPH / Eurographics Symposium on Computer Animation	*2026*
Journal of Computational Physics	*2025, 2026*
Nuclear Engineering and Design	*2025*

Grants

Deutschlandstipendium Germany
Selected based on academic performance and social commitment. *2019 – 2020*

Teaching

Seminars

Advanced Topics in Animation University of Freiburg
2026

Supervised Graduate Students

Julian Karrer University of Freiburg
2026

Joshua Fässler University of Freiburg
Strong Rigid-Fluid Coupling using a Unified Implicit SPH Solver *2024 – 2025*

Philipp Sandweg University of Freiburg
Implicit SPH Simulation of Elastic Solids in a Multi-Material Framework *2023 – 2024*

Andrés Sandoval University of Freiburg
Global Matrix-free Pressure Solver for Performant SPH Fluid Simulations *2022 – 2023*

Supervised Undergraduate Students

Johann-Frederik Isele University of Freiburg
State Equation SPH Fluid Simulation *2026*

Marius Hägele University of Freiburg
Implementation and evaluation of a 2D state equation SPH fluid solver *2025 – 2026*

Georgij Hergenräder University of Freiburg
Smoothed Particle Hydrodynamics Fluid Simulation in 2D - Implementation and Evaluation *2025 – 2026*

Skills

Languages: Fluent in English and German (written, verbal, and reading)
Coding: C++, Python, C, C#, OpenCL, SYCL, CUDA
Frameworks: NumPy, Godot, Unreal Engine, Blender